

E-paper displays use technology enabled by electronic ink, which carries a charge, enabling it to be updated through electronics. Electronic ink is ideally suited for display as it is a reflective technology that requires no front or back light, is viewable under a wide range of lighting conditions, including direct sunlight, and requires no power to maintain an image

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E-paper was first developed in the 1970s by Nick Sheridon at Xerox's Palo Alto Research Center (USA). The first e-paper—Gyricon—consisted of polyethylene spheres between 20 and 100 micrometres. Each sphere was composed of negatively charged black plastic on one side and positively charged white plastic on the other. The spheres were embedded in a transparent silicone sheet, with each sphere suspended in a bubble of oil so that they could rotate freely. The polarity of the voltage applied to each pair of electrodes then determined whether the white or black side was the face-up, thus giving the pixel a white or black appearance.

E-paper has also been produced using technologies such as organic light emitting diode (OLEDs). Other research efforts have involved using organic transistors embedded into flexible substrates, including attempts to build them into conventional papers. Simple coloured e-papers consist of a thin coloured optical filter added to the monochrome technology described above. The array of pixels is divided into triads, typically consisting of the standard cyan, magenta and yellow, in the same way as CRT monitors (which use subtractive primary colours as opposed to additive primary colours). The display is then controlled like any other electronic colour display.

OLED technology, pioneered and

patented by Kodak/Sanyo, enables full-colour, full-motion flat panel displays with a level of brightness and sharpness not possible with other technologies. Unlike traditional LCDs, OLEDs are self-luminous and do not require backlighting, diffusers, polarisers, or any of the other baggage that goes with LCDs. Essentially, OLED consists of two charged electrodes sandwiched on top of some organic light emitting materials. This eliminates the need for bulky and environmentally undesirable mercury lamps and yields a thinner, more versatile and compact display. Their low power consumption provides for maximum efficiency and helps minimise heat and electric interference in electronic devices. Armed with these features, OLED displays communicate more information in a more engaging way while adding less weight and taking up less space. There are two forms of OLED displays: passive-matrix and active-matrix.

The basic OLED cell structure consists of a stack of thin organic layers sandwiched between a transparent anode and a metallic cathode. The organic layers comprise a hole-injection layer, a hole-transport layer, an emissive layer and an electron-transport layer. When an appropriate voltage (typically between 2 and 10 volts) is applied to the cell, the injected positive and negative charges recombine in the emissive layer to produce light (electro luminescence). The structure of the organic layers and the choice of anode and cathode are designed to maximise the recombination process in the emissive layer, thus maximising the light output from the OLED device.

OLEDs have significant advantages like a robust design, a viewing angle up to 160 degrees, high resolution, thin as a dime, production advantages, video capabilities, lighter in weight, faster and use less power. There still remain some engineering hurdles that prevent OLEDs from performing at their optimum levels.

Applications

E-paper applications include e-book readers capable of displaying digital versions of books and e-paper magazines, newspapers, electronic pricing labels in retail shops, time-tables at bus stations, digital photo frames, displays embedded in smart cards, cellphone displays and electronic billboards.

Current status of technology

Great progress has been made in the field of e-paper since its invention. Companies such as E Ink, SiPix and Polymervision, as well as giants like Sony, IBM, Hewlett-Packard, Philips, Fujitsu, Hitachi, Siemens, Epson and many others, are continuing to develop e-paper technology. However, e-paper still lacks good colour reproduction. The contrast ratio in common devices could be described as similar to dirty newspaper, though newly-developed implementations are slightly better.

E-paper technologies have a very low refresh rate compared to LCD technologies. This prevents producers from implementing sophisticated interactive applications (using fast moving menus, mouse pointers or scrolling) like those possible on handheld computers. This has a major impact on the e-paper capacity to display zoomed versions of large text or images on a small display. At this stage, some of the products based on the technology are little more than expensive gimmicks but in the near future, there will be products with substantial applications of e-paper display technology. ●

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